Separate Instruction and Data Spaces

1 Core Idea

Definition

Separate instruction and data spaces means a process has one virtual address space for program instructions and another virtual address space for data.

The instruction space is often called **I-space**. The data space is called **D-space**.

Main Benefit

$\boxed{\text{separate I-space and D-space can double the usable virtual address range}}$

2 Single Address Space

Most computers use one address space for both program text and data.

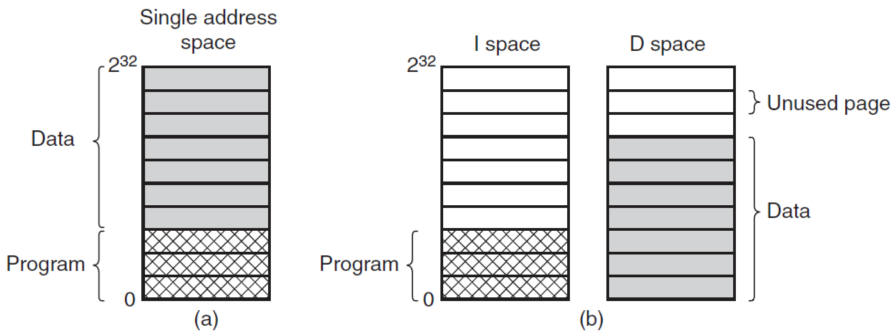

Figure 3-24. (a) One address space. (b) Separate I and D spaces.

In panel (a), the program and data share one address space.

Part	Meaning
Program	Machine instructions, also called program text.
Data	Variables, heap, stack, and other data memory.
Single limit	Program and data together must fit in the same address range.

Problem

If the single address space is small, the program and data compete for the same limited virtual addresses.

3 Separate I-Space and D-Space

Panel (*b*) shows separate spaces:

- **I-space**: holds instructions;
- **D-space**: holds data.

Each space starts at virtual address 0 and runs up to its own maximum.

Key Difference

Address 0 in I-space is not the same as address 0 in D-space. They belong to different virtual address spaces.

4 Why This Helps

With one address space, program and data must share the same range.

With separate spaces:

$$\text{program can use a full I-space}$$

$$\text{data can use a full D-space}$$

Design	Effect
Single address space	Program and data together must fit in one space.
Separate I/D spaces	Program and data each get their own space.

This was especially useful on machines with small address sizes, such as 16-bit systems.

5 Linker Requirement

The linker must know whether separate I-space and D-space are being used.

Why the Linker Cares

In a single address space, data usually starts after the program. With separate I/D spaces, data can be relocated to virtual address 0 in D-space.

So the layout changes:

Mode	Data location
Single address space	Data is placed after program text.
Separate I/D spaces	Data starts at virtual address 0 in D-space.

6 Paging with Separate Spaces

Both I-space and D-space can be paged independently.

Address type	Page table used
Instruction fetch	I-space page table.
Data read or write	D-space page table.

Hardware Role

The hardware knows whether it is fetching an instruction or accessing data, so it can choose the correct page table.

7 OS Implications

Separate I/D spaces do not fundamentally complicate paging.

- I-space has its own page table.
- D-space has its own page table.
- Instruction addresses are translated through the I-space table.
- Data addresses are translated through the D-space table.

The main difference is that the OS tracks two mappings instead of one.

8 Modern Relevance

Modern virtual address spaces are usually large, so separate I/D address spaces are less necessary for ordinary process address-space size.

However, the idea still appears in cache design.

L1 Cache

Many processors split the L1 cache into an instruction cache and a data cache.

This is similar in spirit:

- instruction fetches use the instruction cache;
- data loads/stores use the data cache.

Why Still Useful

Even when virtual address spaces are large, L1 cache space is small and performance-critical, so separating instruction and data traffic can help.

9 Final Summary

One-Box Summary

Single address space: program and data share one virtual range.
Separate I/D spaces: instructions and data have separate ranges.
Instruction fetches use the I-space page table.
Data accesses use the D-space page table.
The idea is still common in split L1 instruction/data caches.